The empirical recurrence for OEIS A262422, proved by transfer matrix

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Abstract

OEIS A262422 counts arrays of width 4 over $\{0,1\}$ whose rows, read as base-2 numbers, are not divisible by 3 and whose columns, read the same way, are divisible by 3. The entry carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. A column's value depends on every row of the array, so the condition is not local; but its RESIDUE is built one row at a time by Horner's rule, and the vector of column residues is a state of bounded size. The count is then a walk count on $S = 81$ states.

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1 The conjecture

OEIS A262422 is "Number of (3+1) X (n+1) 0..1 arrays with each row divisible by 3 and column not divisible by 3, read as a binary number with top and left being the most significant bits.". It has offset 1 and begins

$$10, 12, 270, 1260, 15310, 124572, 1299070, 12380940, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 10*a(n-1) +46*a(n-2) -460*a(n-3) -609*a(n-4) +6090*a(n-5) +2164*a(n-6) -21640*a(n-7) -1600*a(n-8) +16000*a(n-9).

As of the "Last modified" line on the live entry (Mar 15 2023 11:09:13, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 4 cells across and entries $x(i, j) \in \{0, 1\}$, the digits of base 2. Row i of an array with R rows is the number

$$\rho_i \ = \ \sum_{j=1}^{4} x(i,j)\, 2^{4-j},$$

the LEFT digit being the most significant, and column j is

$$\gamma_j \ = \ \sum_{i=1}^{R} x(i,j)\, 2^{R-i},$$

the TOP digit being the most significant. The entry asks that every ρ_i be not divisible by 3 and every γ_j be divisible by 3. The entry writes the array with n as the number of COLUMNS. The walk is run down the transpose, which exchanges the row condition with the column condition; both readings put the FIRST digit in the most significant place, so nothing else changes under that exchange.

3 Column residues by Horner's rule

The row condition is a condition on one row and is imposed as each row is chosen. The column condition is not local: γ_j depends on every row. What is local is its residue.

Lemma 1. *Let $\gamma_j^{(i)}$ be the value of column j of the first i rows, read the same way. Then $\gamma_j^{(0)} = 0$ and*

$$\gamma_j^{(i)} \;=\; 2\,\gamma_j^{(i-1)} + x(i,j), \qquad \textit{hence} \qquad \gamma_j^{(i)} \bmod 3 \;=\; \big(2\,(\gamma_j^{(i-1)} \bmod 3) + x(i,j)\big) \bmod 3.$$

Proof. Adding a row at the BOTTOM shifts every earlier digit one place up in significance, which is multiplication by 2, and puts the new digit in the units place. Reduction modulo 3 commutes with both operations. ☐

So the vector $\big(\gamma_1 \bmod 3, \dots, \gamma_4 \bmod 3\big)$ is a state: at most 3^4 of them, and each array row that satisfies the row condition carries one such vector to another. The walk starts at the all-zero vector, which is the empty array, and the END vector picks out the residue vectors in which every entry is divisible by 3 — so the end vector is not the all-ones vector, and the column condition is imposed once, at the end, exactly as the entry states it. With M the adjacency matrix,

$$a(n) \;=\; \iota^{\top} M^{\,j} \tau, \qquad S = 81.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+9} - \sum_i c_i M^{j+9-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. ☐

Theorem 3.

$$a(n) \;=\; 10\,a(n-1) + 46\,a(n-2) - 460\,a(n-3) - 609\,a(n-4) + 6090\,a(n-5) + 2164\,a(n-6) - 21640\,a(n-7) - 1600\,a(n$$

for every $n > 7$.

Proof. The vector $w = q(M)\tau$ was formed by 9 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 7 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. ☐

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array of that size, turn each row and each column into a base-2 number with the first digit most significant, and test the divisibility directly. Neither Horner's rule nor any residue state is involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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